

Student 24 **MALLI CHANDANA** 192472250RD

5 / 5 submitted

### Q1. Program based on Classes and Objects

Score: 3.33 TC: 2/3 passed

#### CODE

```
import java.util.*;

class Student{
    int rollNo;
    String name;
    double marks;

    void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        Student s = new Student();

        s.rollNo = sc.nextInt();
        sc.nextLine();
        s.name = sc.nextLine();
        s.marks = sc.nextDouble();

        s.displayDetails();

        sc.close();
    }
}
```

#### INPUT

```
101
rose
97.9000
```

#### OUTPUT

```
Roll No: 101
Name: rose
Marks: 97.9
```

## Q2. Initialize Object Data Using Setter Methods

Score: 5.00 TC: 3/3 passed

### CODE

```
import java.util.Scanner;
public class Student {
    private int rollNo;
    private String name;
    private double marks;
    public void setRollNo(int rollNo)
    {
        this.rollNo = rollNo;
    }
    public void setName(String name){
        this.name = name;
    }
    public void setMarks(double marks){
        this.marks = marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name );
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());
        s.displayDetails();
        sc.close();
    }
}
```

### INPUT

77  
chandana  
98.0

### OUTPUT

Roll No: 77  
Name: chandana  
Marks: 98.0

### Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

#### CODE

```
import java.util.Scanner;
public class Student {
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo = 101;
        name = "Rahul";
        marks = 85.5;
    }
    Student (int rollNo, String name, double marks){
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }
    void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        Student s1 = new Student();
        s1.displayDetails();
        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        Student s2 = new Student(rollNo, name, marks);
        s2.displayDetails();
    }
}
```

#### INPUT

```
77
chandana
97.6
```

#### OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 77
Name: chandana
Marks: 97.6
```

#### Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

##### CODE

```
import java.util.Scanner;
public class Employee{
    int empId;
    String empName;
    double basicSalary;
    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
    public static void main(String[] args ){
        Scanner sc = new Scanner(System.in);
        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();
        Salary s = new Salary();
        s.getEmployeeDetails(empId,empName,basicSalary);
        s.calculateSalary();
        s.displayDetails();
        sc.close();
    }
}
class Salary extends Employee{
    double hra,da,netSalary;
    void calculateSalary(){
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        netSalary = basicSalary + hra + da;
    }
    void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Net Salary: " + netSalary);
    }
}
```

##### INPUT

```
100
chandana
70000
```

##### OUTPUT

```
employee id: 100
employee name chandana
basic salary 70000.0
HRA: 14000.0
da: 7000.0
net salary: 91000.0
```

## Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

### CODE

```
import java.util.Scanner;
public class Employee {
    int empId;
    String empName;
    void getEmployeeDetails(int empId,String empName){
        this.empId = empId;
        this.empName = empName;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();
        GrossSalary g = new GrossSalary();
        g.getEmployeeDetails(empId,empName);
        g.getBasicSalary(basicSalary);
        g.calculateSalary();
        g.displayDetails();
        sc.close();
    }
}
class Salary extends Employee {
    double basicSalary;
    void getBasicSalary(double basicSalary){
        this.basicSalary = basicSalary;
    }
}
class GrossSalary extends Salary {
    double hra,da,grossSalary;
    void calculateSalary(){
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        grossSalary = basicSalary + hra + da;
    }
    void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra );
        System.out.println("DA: " + da);
        System.out.println("Gross Salary: " + grossSalary);
    }
}
```

### INPUT

101  
Rahul  
50000

### OUTPUT

Employee ID: 101  
Employee Name: Rahul  
Basic Salary: 50000.0  
HRA: 10000.0  
DA: 5000.0  
Gross Salary: 65000.0